



An integrated design process for mission optimized design of Unmanned Aerial Vehicles

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An integrated numerical design process for conceptual and preliminary design of unmanned aerial vehicles tailored for optimal fulfillment of predefined mission tasks will be introduced. A numerical multidisciplinary aircraft sizing method is implemented for creating consistent designs of flying platforms which are then evaluated by virtually performing sample missions using a mission simulation tool specifically implemented for this task. A genetic algorithm is used to vary geometry and payload sensor parameters in order to create an aircraft design optimally tailored to achieve the best mission performance possible. The method is implemented in a modular design environment providing the ability to use different calculation methods for different disciplines during the design process, e.g. aerodynamics, mass prediction or propulsion system layout.

I. Introduction

The majority of civil application scenarios for Unmanned Aerial Vehicles (UAV) consists of either a mapping task (e.g. crop surveillance) or a search-and-track task (e.g. find missing persons). In contrast to classic design objectives used in aircraft design (e.g. fuel efficiency, speed, range) UAV rely heavily on the interaction with the environment using their onboard sensors.

This necessitates new approaches for mission based design methods that not only emphasize on fuel/energy efficiency for flying a certain mission, but furthermore take into account the system's ability to efficiently fulfill its mission tasks. Therefore sensor imaging results, datalink quality predictions and path planning requirements need to be regarded as optimization goals in the design process besides fuel/energy efficiency. The result of this design process is an aircraft system specifically tailored to the mission requirements. It does not necessarily deliver the most fuel efficient aircraft or make use of the sensor with the highest resolution and best image quality available, but as a complete system fulfill its requirements as good as technically possible.

II. Design process

The integrated design process maximizes a scalar mission performance index by varying aircraft geometry and payload sensor parameters using a genetic optimization algorithm.

The design process consists of a numerical aircraft sizing process, which delivers consistent aircraft designs in a preliminary design stage, and a mission simulation model to evaluate and quantify mission performance of these designs. For its implementation a modular design environment was created in MATLAB enabling efficiently performing all calculation tasks based on one central, consistent dataset.

The genetic optimization algorithm "GAME" by Langer [1] is used to find the best suited aircraft design by maximizing a scalar mission fulfillment index (see section III). "GAME" was chosen due to its ability to handle discrete and continuous design variables simultaneously and due to its robustness in finding global optima during the process of numerical optimization.

Genetic optimization algorithms are inspired by the processes of natural selection. A "population" of a finite number of "individuals" is evaluated by their ability of fulfilling a certain objective. In an iterative process, new generations are created, which themselves are evaluated by their fitness. By repeating this process, properties of the individuals should converge to an optimal solution of the design problem.

In the beginning an initial "population" of individuals is generated from randomized values of the design variables

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(e.g. wingspan, sensor parameters). For each individual represented by a set of design variable values the aircraft sizing process described in section V is executed, and the resulting consistent aircraft designs are evaluated in terms of their mission performance. All resulting individuals are then ranked by their mission performance indices and fulfillment of constraint functions. To create a new "children generation" methods of randomized recombination, mutation, but also preserving best individuals of the parent generation are used. These processes are repeated until convergence to an optimal design is achieved.

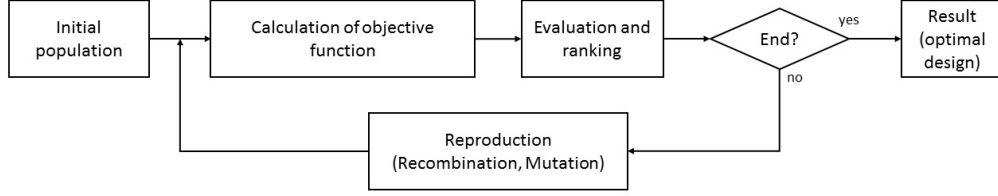


Fig. 1 Procedural diagram for optimized aircraft design process

III. Mission simulation and quantification of fulfillment of mission requirements

One of the main issues to be addressed in the design methodology introduced here is the ability to evaluate and quantify the grade of fulfillment of mission requirements. This is necessary in order to compare different preliminary system architectures consisting of an Unmanned Aerial Vehicle and their respective sensor payloads. Therefore methods to achieve quantifiable mission fulfillment indices are defined and implemented into the simulation tool chain framework to virtually carry out missions [2].

In order to create a suitable method for evaluating mission performance incorporated in a multidisciplinary design process the following requirements have to be fulfilled:

Scalar: Mission fulfillment should be quantified by a scalar mission fulfillment index to be used as the result of an objective function for numerical optimization.

Normalization: In order to facilitate comparability between results of mission evaluation each contributing parameter should be normalized by a reference value. Also, by normalization dominance of parameters with large values due to their physical units is avoided.

Unitless: Quantification of mission fulfillment should be done with dimensionless parameters. This can be achieved by normalization with reference values of the same physical unit.

Weighting: Each contributing effect should be weighted by its importance by multiplying an importance factor. To facilitate formulation of an objective function for optimization and to avoid Pareto optimal problem formulations mission success is condensed into a scalar mission performance index M . In order to take into account all contributing factors a "weighted-sum approach" is used, where degradation factors are multiplied to the corresponding effectivity parameters:

$$M = \sum_i \alpha_i \cdot \frac{Effectivity_i \cdot Degradation_i}{Effectivity_{base,i}} + \sum_j \beta_j \cdot \frac{Effort_{base,j}}{Effort_j} \quad (1)$$

The formulation of metrics is still a topic of ongoing research. Therefore, other methods for formulation of mission performance metrics are under investigation.

Parameters used in this method are classified in three main categories:

Effectivity parameters: Parameters positively influencing mission success, e.g. covered target area, number of discovered targets.

Effort parameters: Parameters quantifying what needs to be invested to carry out a mission, e.g. fuel burned, time required.

Degradation factors: Factors degrading mission success by not completely fulfilling requirements, e.g. reduced ground resolution of imaging sensors, obstacles in the line of sight.

Each contributing effect is weighted by its importance by multiplying a weighting factor α_i , β_i . The sum of the weighting factors equals unity ($\sum \alpha_i + \sum \beta_j = 1$). The exact definition of the mission fulfillment index needs to be adapted to the requirements of the mission to be analyzed. For example for search and rescue missions the most

important criteria are time until detection of target(s) and searching time to completely cover the target area. Criteria like fuel consumption and cost get second priority. In contrast, for agricultural crop monitoring first order priorities are fuel cost, while mission time becomes less important.

IV. Implementation of a mission simulation model

The mission simulation algorithm is implemented as a control flow model in Simulink. It consists of three major components, depicting a basic architecture of modern UAV avionics. Due to the modular architecture based on separate interconnected model blocks it can be easily adapted to different kinds of missions, for use of different types of payload sensors or to provide further calculation results.

A "Mission Management Model" acts as central knot to all other components. It is responsible for generating flight path commands, distributing flight state information such as position and attitude to the sensor model and storing simulation results to the aircraft data model for further evaluation after the mission simulation process.

The "Flight Control Model" calculates aerodynamic and propulsion forces and integrates the linearized equations of motion to determine the aircraft's current position, velocity and attitude angles while neglecting its rotational dynamics. The "Sensor payload" model calculates sensor results depending on the type of task of the current mission.

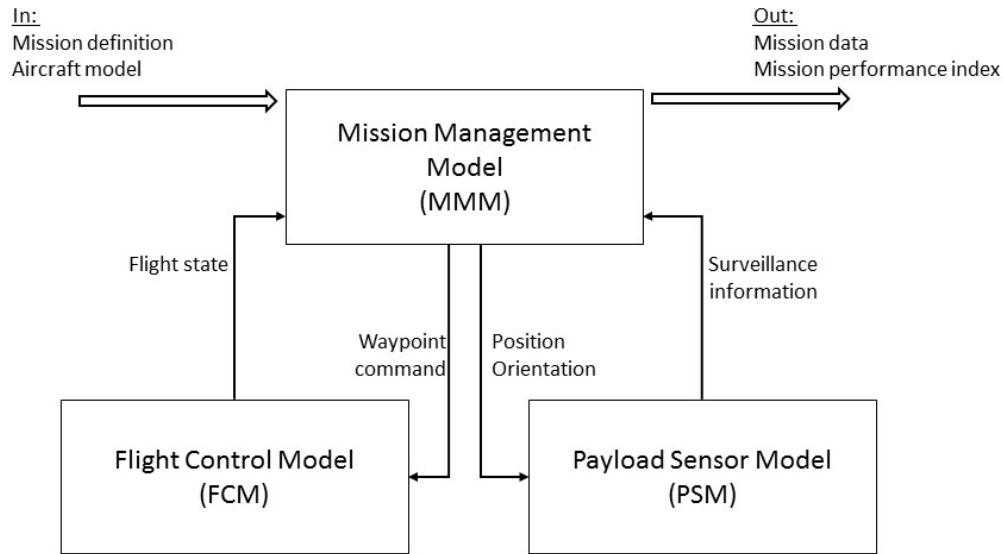


Fig. 2 Architecture of mission simulation algorithm

A. Mission Management Model

The Mission Management Model element works as central control element of the simulation model. It distributes required input data for other model elements. Also it is responsible for generating waypoint navigation commands to be provided to the Flight Control Model element. The Mission Management Model element also performs selection of data to be provided for processing after the execution of the simulation model.

B. Flight Control Model implementation

The FCM is implemented in a simplified way to give a reasonable estimation of the aircraft's current flight state at each time step. In order to reduce complexity while maintaining sufficient model fidelity rotational dynamics, wind influence and aerodynamic sideslip are neglected. At first angle of attack α , bank angle μ and thrust lever setting δ_T are determined by proportional-integral-differential feedback of the difference between commanded and current flight state. Next all forces acting on the aircraft are calculated. Lift coefficient C_L and drag coefficient C_D are interpolated from trimmed aerodynamic polar data provided by the central data model. The polar data is generated as a two-dimensional matrix depending on angle of attack α and altitude h . Similarly thrust forces are interpolated from a three-dimensional

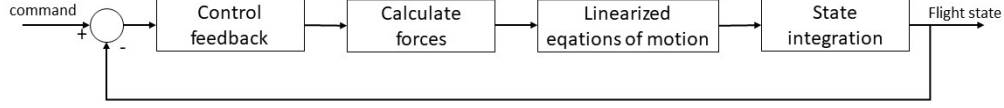


Fig. 3 Working scheme of flight control model

matrix as a function of throttle setting δ_T , airspeed V and altitude h . If a combustion engine is used current aircraft mass is determined by integration of used fuel. For electric propulsion systems used energy is determined. These forces are used to solve the linearized equations of motion in polar coordinate form:

$$\dot{V} = \frac{T - D}{m} - 9.81 \cdot \sin \gamma \quad (2)$$

$$\dot{\chi} = \frac{L \cdot \sin \mu}{m \cdot V \cdot \cos \gamma} \quad (3)$$

$$\dot{\gamma} = \frac{L \cdot \cos \mu}{V \cdot m} - \frac{9.81 \cdot \cos \gamma}{V} \quad (4)$$

By integrating equations (2)-(4) and neglecting wind influence the aircraft's velocity is determined in a north-east-down coordinate frame.

Due to neglecting wind and sideslip attitude angles are simplified to:

$$\Psi = \chi \quad (5)$$

$$\Theta = \alpha + \gamma \quad (6)$$

$$\Phi = \mu \quad (7)$$

$$(8)$$

Position at each time step is calculated by integrating the position differential equations in WGS84-coordinates.

$$\dot{\lambda} = \frac{V \cdot \sin \chi \cdot \cos \gamma}{(N_\mu + h) \cdot \cos \mu} \quad (9)$$

$$\dot{\mu} = \frac{V \cdot \cos \chi \cdot \cos \gamma}{M_\mu + h} \quad (10)$$

$$\dot{h} = V \cdot \sin \gamma \quad (11)$$

with

$$N_\mu = \frac{a}{\sqrt{1 - e^2 \sin^2 \mu}} \quad (12)$$

$$M_\mu = N_\mu \cdot \frac{1 - e^2}{1 - e^2 \sin^2 \mu} \quad (13)$$

$$e = 2f - f^2 \quad (14)$$

$$f = \frac{a - b}{a} \quad (15)$$

N_μ denotes the curvature radius of the earth ellipsoid's main vertical axis, M_μ the meridian curvature radius, e first eccentricity and f flattening. $a = 6378137\text{m}$ and $b = 6356752.3142\text{m}$ are the length of the reference ellipsoids semimajor axis and semiminor axis.

C. Payload Sensor model implementation

The implementation of the sensor model depends on the parameters required to evaluate mission performance. Currently two different models have been implemented: One sensor model to evaluate performance for aerial mapping missions like agricultural crop monitoring. The other model is used to evaluate target detection probabilities of point targets for search and rescue (SAR) missions.

Due to the modular architecture of the mission simulation model other types of sensors and calculation methods can easily be added.

1. Optical sensor model for mapping missions

For the creation of orthomosaic images of an area it is relevant to calculate the width of the optical sensor footprint, the momentary ground sample distance as measure of resolution as well as overlap between images [3][4]. Ground sample distance is calculated by equation (16).

$$GSD = 2 \cdot \tan \frac{FOV_H}{2 \cdot H_{pix}} \cdot h \quad (16)$$

FOV_H is the horizontal field of view, H_{pix} the number of sensor pixels in horizontal direction and h the current altitude over ground of the aircraft.

2. Optical sensor model for Search And Rescue (SAR) missions

For SAR missions the probability of detection of targets in combination with the required search time is relevant for mission success.

To determine a target's probability of detection, recognition and identification the criteria defined by Johnson is used [5][6]. It is a semi-empirical method that calculates the probability of performing the respective task by comparing the number of sensor cycles N covered by a target to a reference value N_{50} for 50% probability.

$$P(N) = \frac{(N/N_{50})^{2,7+0,7 \cdot (N/N_{50})}}{1 + (N/N_{50})^{2,7+0,7 \cdot (N/N_{50})}} \quad (17)$$

In the original publication Johnson specifies reference values N_{50} to be used with 0.75 for detection, 3.0 for recognition and 6.0 for identification. Schmieder and Weathersby introduce an approach for adapting Johnson's criteria to different target scenarios by adapting the respective N_{50} values [7].

D. Results of the mission simulation model

The simulation model is configured to deliver vectors of data for each sample time point for further analysis. Besides data calculated by the sensor models the following data is provided:

- Position in latitude, longitude and altitude in WGS84 coordinates
- Kinematic velocity with speed, course angle and climb angle in North-East-Down coordinates
- Attitude angles with heading, pitch and roll attitude
- Current control variable settings (angle of attack, bank angle, thrust lever setting)
- For combustion engine: Mass, used fuel mass and fuel flow
- For electric propulsion system: Used energy

V. Preliminary aircraft design methods for creation of consistent aircraft data sets

As described in section II, a genetic optimization algorithm is used to find an unmanned aircraft design that optimally fulfills a certain mission task. Therefore, a process to create consistent designs of flying platforms is required. Such a dataset contains information about general geometric data, masses, aerodynamic polars and propulsion system data. Semiempirical textbook methods for aircraft design are often limited in their applicability to small UAV configurations. Therefore, simplified physical methods, especially for aerodynamics and structural weight estimation, are used where applicable. By this approach accuracy of calculation results is improved.

Figure 4 gives an overview of the design process. First for a given wing geometry an aerodynamic polar covering the expected flight envelope is calculated, then the wing's mass is estimated by creating a simplified structural layout. For the current mass the propulsion system is designed and fuselage geometry is determined. Also, tail surfaces are sized. Then masses of all elements are determined to update the configuration's overall mass. With the updated mass this process is iterated until masses converge.

Special consideration is given to fuselage layout and sensor placement to take obscuration of field of view into account. The result is a feasible design that is able to perform mission requirements for the design process.

By systematically varying initial settings of this process (e.g. initial wing geometry) multiple designs are created and evaluated. Systematic variation of settings is performed by a genetic optimization algorithm, until no further improvement in terms of a scalar mission performance index is achieved.

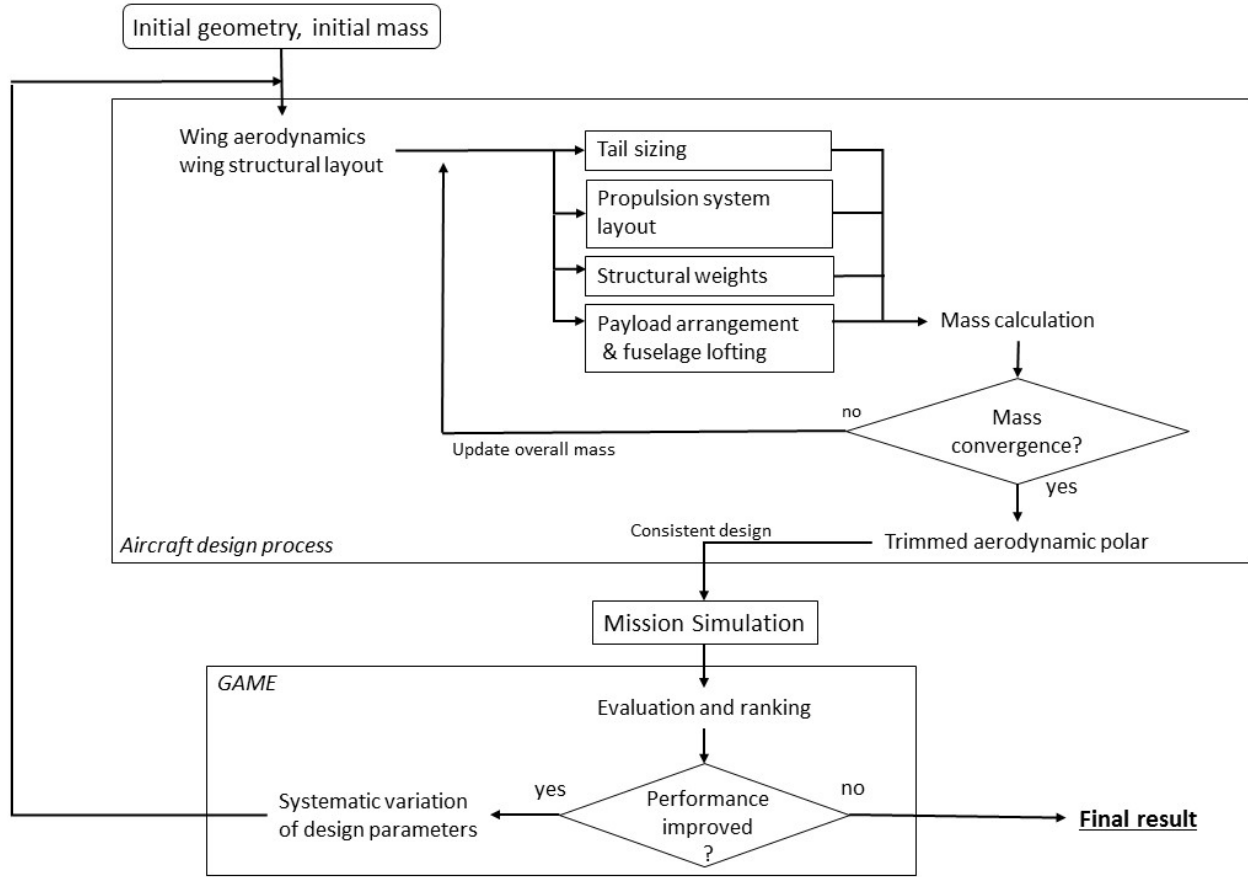


Fig. 4 Design process for calculation of a consistent aircraft dataset

A. Aerodynamic calculations

For all calculations of aerodynamic behavior of the aircraft the program "PAWAT" is used [8]. PAWAT is based on three-dimensional lifting line theory and is capable of including nonlinear airfoil data to calculate aerodynamic forces and moments as well as stability derivatives.

In the aircraft design process presented in this publication PAWAT is used calculate distributions of lift and structural loads in spanwise direction of lifting surfaces as well as untrimmed and trimmed polars of combinations of multiple lifting surfaces.

To take pressure and surface friction drag of fuselages and other bodies and interference drag into account a semiempirical approach is used [9].

1. Untrimmed polar

PAWAT is used to generate untrimmed aerodynamic polars. The polar consists of lift coefficient c_L , drag coefficient c_D and pitch moment coefficient c_M . All these coefficients are calculated depending on aerodynamic speed v_A , altitude h , angle of attack α and trim surface deflection angle η_h .

2. Calculation of trimmed polar

Trimmed aerodynamic polar data consist of all possible level flight states without acceleration in airspeed, loss or increase in altitude or turning flight. A trim state in longitudinal flight is defined here as an equilibrium in lift and

gravity force and in pitching moment.

$$m \cdot g - \frac{\rho}{2} v^2 \cdot c_L \cdot S_{ref} = 0 \quad (18)$$

$$\frac{\rho}{2} v^2 \cdot c_M \cdot c_{MAC} \cdot S_{ref} = 0 \quad (19)$$

Trimmed flight states are extracted from untrimmed polar data by interpolating $c_M = 0$ and required lift coefficient value for each desired trim point.

Trimmed polar data consist of lift coefficient $c_{L,trim}$, drag coefficient $C_{D,trim}$, trim speed v_{trim} and required trim surface deflection η_{trim} . All coefficients are provided as a function of angle of attack α and altitude h .

B. Wing weight estimation

Since common weight estimation methods for wing structures usually cannot be applied to unmanned aircraft a simplified method for layout of fiber composite structures is used. The method was developed by Christian Roessler [10].

1. Lift distribution and wing loads

For preliminary structural layout of lifting surfaces of small unmanned aircraft the distribution of bending moment and torsional moment in spanwise direction are critical. Other loads on wing structures are not taken into account for weight estimation of wing and tail surfaces [11].

PAWAT is used to calculate lift and torsional moment distribution using a lifting line method. Lift force and pitching moment at each panel's reference point are then integrated to calculate structural loads [1].

Dimensioning aerodynamic loads are predefined requirements for the aircraft sizing process by providing a maximum design speed with a corresponding load factor n_{design} . On all loads a safety factor of 1.5 is applied.

2. Structural sizing of CFRP sandwich structures

Fiber composite structures for small UAVs typically consist of a sandwich shell structure enforced by a wing spar (figure 5). It is assumed that all bending loads are carried by the wingspar, while all torsional loads are carried by the wing shell.

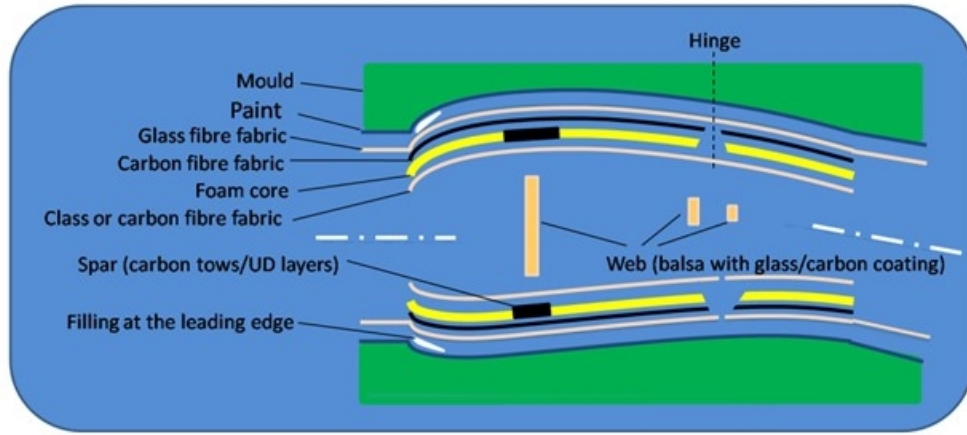


Fig. 5 Schematic structural layout of composite wing [10]

To calculate thickness of the fiber layer of the wing shell the method by Dolch is used [11]. The required torsional stiffness at the wing root section can be determined by assuming that a surrogate structure with the same torsional constant may not be twisted by more than $\phi = 4^\circ$ to avoid flutter.

$$I_{root} = \frac{M_{t,root} \cdot b}{2 \cdot \phi \cdot G} \quad (20)$$

$M_{t,root}$ is the torsional moment at the root section, b the wingspan and G the shear modulus of the outer shell. The required torsional constant is reduced in spanwise direction proportionally to the torsional moment distribution.

$$I(y) = I_{root} \cdot \frac{M_{t,root}}{M_t(y)} \quad (21)$$

$$t(y) = \frac{I(y) \cdot U(y)}{4A(y)^2} \quad (22)$$

$t(y)$ is the fiber layer thickness at spanwise coordinate y , U is the airfoil's circumference and A the wing section's area. The wingspar consists of two spar caps carrying all bending stress of the wing. Their thickness is set equal to the thickness of the shell's foam core. The required width of the spar caps can be solved to:

$$w_{spar}(y) = \frac{6 \cdot M_b(y) \cdot H}{\sigma_b \cdot (H^3 - h^3)} \quad (23)$$

$M_b(y)$ is the bending moment and σ_b the material's maximum bending strength, H the local thickness of the wingspar and h the inner height of the shear web between the spar caps.

The shear web consists of a balsa wood core covered with glass fiber laminate. The areal weight of the glass laminate is calculated by:

$$m_{web} = \frac{Q(y)}{k_\tau \cdot h} \quad (24)$$

$Q(y)$ is the shear force and $k_\tau \approx 10000m$ a material constant for glass fiber laminate.

Additional weights are added for bonding of the shells and for paint.

All weights of structural components are summed up to provide an estimation of the weight of the wing structure.

C. Payload sensor placement and fuselage lofting

A fuselage is required for housing an arbitrary number of system components as well as a sensor payload and propulsion system components. The method used therefore needs to create sufficient space to hold all components while maintaining a minimal surface area to minimize structural weight and drag. Special consideration is given to placement of an electro-optical sensor payload to maintain an acceptable level of obscuration of its field of regard by structural components of the aircraft like wings and tail surfaces. All components and fuselage hull are arranged so, that a required center of gravity position of the aircraft is maintained to ensure stable flight characteristics.

This method delivers the lightest fuselage hull and component arrangement that maintains a required position of the center of gravity. It also maintains an acceptable percentage of obstruction of a sensor's field of regard, which is defined by the user of the process according to requirements of the mission.

1. Component alignment and creation of fuselage hull

In order to decrease required time for calculation, the design space is restricted to an index list of all components to be fitted into the fuselage. This list contains all possible combinations of order of component numbers. Following this order of index numbers, components are aligned into partitions along the longitudinal axis of the fuselage. In each of these partitions the components are arranged besides and on top of each other until maximal dimensions of the current partition are reached. This arrangement process is repeated by creating further partitions in longitudinal direction until all components are fitted into an order of partitions. All components' positions are then shifted along the longitudinal fuselage axis to adjust the center of gravity to a position required for stable flight.

After component arrangement the fuselage geometry is determined by encasing all components into a tightly fitted hull represented by ellipsoidal cross sections. Component positioning and creation of the fuselage hull is then iterated until the required position of the center of gravity is reached.

Structural weight is estimated using a semiempirical method by multiplying the surface area of the fuselage hull by a mass factor f_{Hull} [10]. Weight for wiring and mounting brackets is accounted for by multiplying the mass of components by the sum of factors f_{wiring} and $f_{brackets}$ [6]. Typical values are $f_{Hull} = 0.75 \frac{kg}{m^2}$, $f_{wiring} = 0.25$ and $f_{brackets} = 0.1$.

$$m_{fuselage} = f_{Hull} \cdot S_{Hull} + (f_{wiring} + f_{brackets}) \sum_i m_{component,i} \quad (25)$$

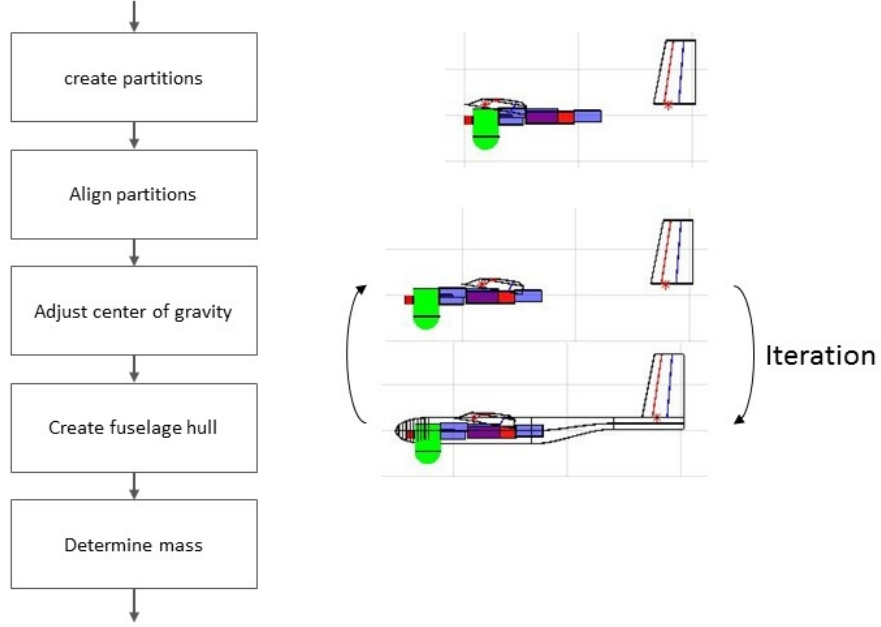


Fig. 6 Schematic overview of algorithm for fuselage creation

2. Obscuration of field of regard

To determine the ratio of interference of structural parts of the aircraft with the optical sensor's field of regard a visibility algorithm is used [12]. The field of regard of the sensor is discretized by a finite number of "rays" represented by angular directions in tilt and pan. Each of these rays is tested for intersection with the aircraft's geometry. Aircraft geometry is discretized to a finite number of triangular facets similar to the definition of the STL geometry data format [13]. Each facet is defined by coordinates of the corner points P_i and its normal vector $\vec{n}_i$ which faces towards the outside of the geometry.

$$P_i = \begin{Bmatrix} p_{1x} & p_{1y} & p_{1z} \\ p_{2x} & p_{2y} & p_{2z} \\ p_{3x} & p_{3y} & p_{3z} \end{Bmatrix} \quad (26)$$

$$\vec{n}_i = \begin{Bmatrix} n_x & n_y & n_z \end{Bmatrix} \quad (27)$$

In order to reduce computation time the aircraft geometry is simplified to planar surfaces for lifting surfaces and trapezoidal cross sections for bodies (see fig. 7).

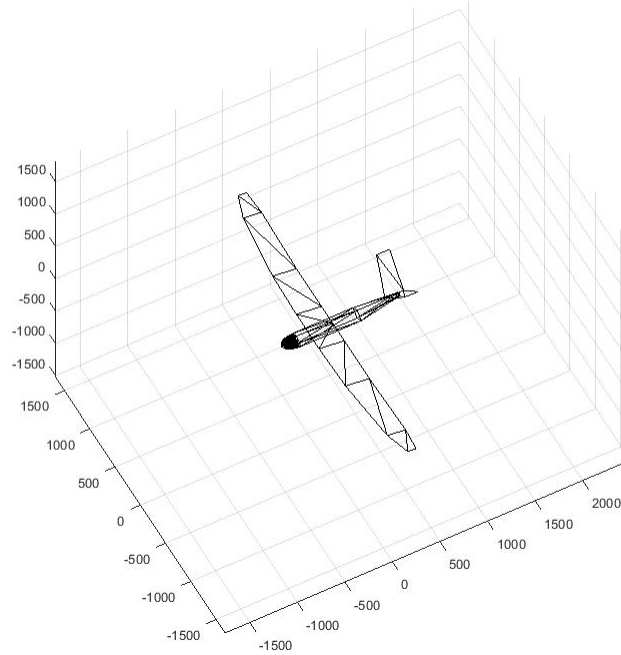


Fig. 7 Simplified geometrical representation using triangular facets [14].

For testing if intersection between one optical ray and one triangular facet of the aircraft structure occurs the Pluecker-representation of these vectors is used. The sign of the Pluecker scalar product of two vectors provides information of the rotational direction of one vector about the other. If all edge vectors $\vec{K}_i$ of a triangular facet show the same rotational direction about a ray vector $\vec{L}$ intersection occurs.

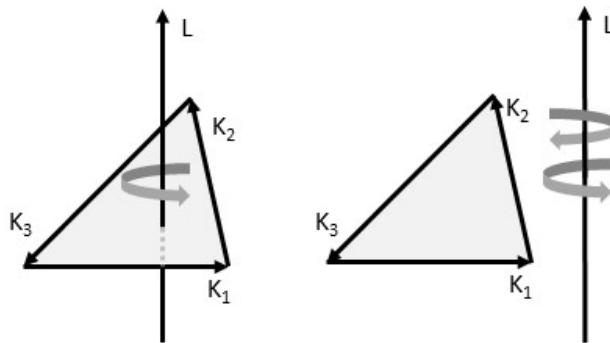


Fig. 8 Illustration of equal/different rotational sense of ray and edge vectors [14].

The Pluecker-representation $\vec{L}$ of a ray vector $\vec{L}$ is defined by:

$$\vec{L} = \begin{pmatrix} a_x b_y & - & b_x a_y \\ a_x b_z & - & b_x a_z \\ a_y & - & b_y \\ a_y b_z & - & b_y a_z \\ a_z & - & b_z \\ b_y & - & a_y \end{pmatrix} \quad (28)$$

$\vec{a}$ and $\vec{b}$ are coordinate vectors of two points on $\vec{L}$. The Pluecker scalar product is defined by:

$$\begin{aligned} \tilde{K}_i \cdot \tilde{L} = & \tilde{K}_1 \cdot \tilde{L}_5 + \tilde{K}_2 \cdot \tilde{L}_6 + \tilde{K}_3 \cdot \tilde{L}_4 + \\ & + \tilde{K}_4 \cdot \tilde{L}_3 + \tilde{K}_5 \cdot \tilde{L}_1 + \tilde{K}_6 \cdot \tilde{L}_2 \end{aligned} \quad (29)$$

To create information about obstruction of field of regard each discrete ray has to be tested against all triangular facets of the aircraft.

The result of this method is shown in figure 9. The Mercator plot shows the projection of aircraft structures in the field of regard of the sensor.

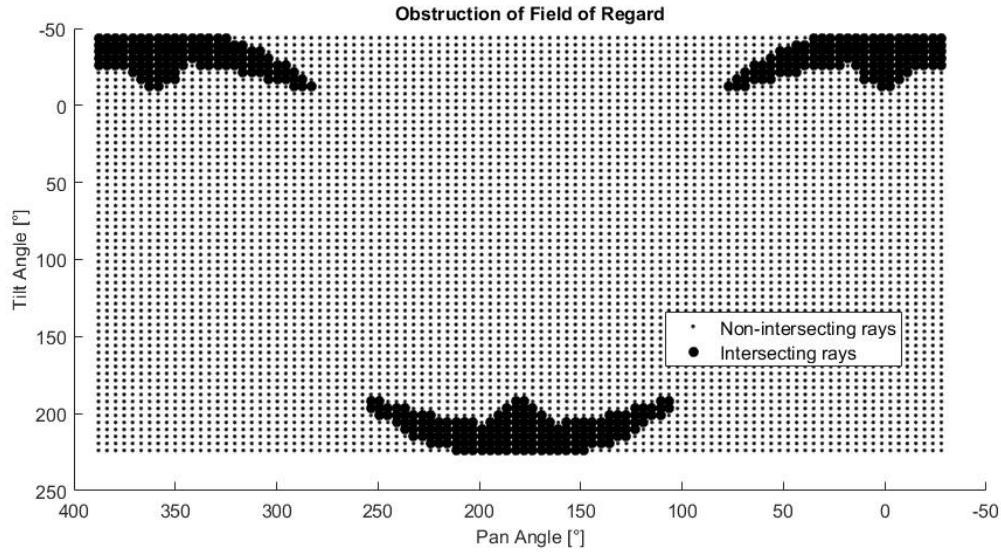


Fig. 9 Mercator plot of optical sensor field of regard. Interference of aircraft structures is visible at top and bottom of the diagram [14]

D. Electrical propulsion system layout

A simple semi-empirical method is used for estimating weight and volume of electrical propulsion system components.

The propulsion system consists of a propeller, an electric motor, an electric speed controller (ESC) and a lithium polymer battery. First thrust and energy requirements as well as design airspeed and altitude are derived from waypoint and speed commands for missions to be simulated. Required thrust is defined by the steepest climb segment to be flown during the design mission. The climb angle is limited to user-defined boundaries $\gamma_{min} \leq \gamma \leq \gamma_{max}$.

$$T_{design} = m \cdot g \cdot \sin \gamma + \frac{\rho}{2} \cdot V_{climb}^2 \cdot C_D \cdot S_{ref} \quad (30)$$

After analyzing design missions for propulsion system requirements the required shaft power and mission energy are determined:

$$P_{electrical} = f_{safety,P} \cdot \frac{T_{design} \cdot v_{design}}{\eta_{prop} \cdot \eta_{mot}} \quad (31)$$

$$E_{bat} = f_{safety,E} \cdot \frac{\sum_i T_i \cdot v_i \cdot t_i}{\eta_{prop} \cdot \eta_{mot} \cdot \eta_{ESC}} \quad (32)$$

T_{design} and v_{design} are required thrust and speed for propulsion system layout, $\eta_{prop}, \eta_{motor}$ and η_{ESC} are efficiency factors of the respective component. $f_{safety,P}$ and $f_{safety,E}$ are additional factors to take installation effects, uncertainty in modeling data and reserves into account.

Statistical data extracted from a database shows proportional relation between motor and ESC weight and required electrical power. The results are shown in table 1.

Component	Specific weight	Specific volume
Motor	$2.7 \cdot 10^{-4} \frac{kg}{W}$	$4000 \frac{kg}{m^3}$
ESC	$1.0 \cdot 10^{-4} \frac{kg}{W}$	$2700 \frac{kg}{m^3}$
Battery	$135 \frac{Wh}{kg}$	$2200 \frac{kg}{m^3}$

Table 1 Reference data for estimation of component weight of the electric propulsion system

E. Tail sizing

1. Determining required position of center of gravity

For small unmanned aircraft it is assumed that an efficient horizontal tail does not provide lift (negative or positive) in its design flight state. To attain an (unstable) equilibrium in pitch the wing needs to compensate its zero lift pitch moment by its lift and its moment arm between wing neutral point and center of gravity.

$$\begin{aligned} \sum M^G &= 0 \\ &= \bar{q} \cdot c_{m0,w} \cdot S \cdot \bar{c} + \bar{q} \cdot c_{l,w} \cdot S \cdot \bar{x}^{GN_w} + \underbrace{\bar{q} \cdot c_{l,ht} \cdot S_{ht} \cdot \bar{x}^{GN_{ht}}}_{=0, \text{ for } c_{l,ht}=0} \end{aligned} \quad (33)$$

Equation (33) can be solved for required center of gravity position relative to the wing neutral point.

$$x^{GN_w} = -\frac{c_{m0,w}}{c_{L,design}} \cdot c_{MAC} \quad (34)$$

$c_{m0,w}$ is the wings zero lift pitch moment coefficient, $c_{L,design}$ the lift coefficient of the design flight state and c_{MAC} is the wing mean aerodynamic chord.

2. Length of tail moment arm

For stable flight a static margin of $SM = 0.12...0.18$ has to be attained. The static stability margin for longitudinal flight is defined by:

$$SM = -\frac{c_{m\alpha}}{c_{L\alpha}} \quad (35)$$

Pitch moment derivative $c_{m\alpha}$ and lift derivative $c_{L\alpha}$ are defined by:

$$c_{m\alpha} = c_{L\alpha,w} \cdot \frac{x^{GN_w}}{c_{MAC}} + \left(1 - \frac{\partial \epsilon_{ht}}{\partial \alpha_w}\right) \cdot c_{L\alpha,ht} \cdot \underbrace{\frac{S_{ht}}{S_{ref}} \cdot \frac{x^{GN_{ht}}}{c_{MAC}}}_{=\bar{V}_{ht}} \quad (36)$$

$$c_{L\alpha} = c_{L\alpha,w} + \left(1 - \frac{\partial \epsilon_{ht}}{\partial \alpha_w}\right) \cdot c_{L\alpha,ht} \cdot \frac{S_{ht}}{S_{ref}} \quad (37)$$

$\bar{V}_{ht} = \frac{S_{ht}}{S_{ref}} \cdot \frac{x^{GN_{ht}}}{c_{MAC}}$ is the tail volume coefficient consisting of the fraction of tail surface are S_{ht} and wing reference area S_{ref} multiplied by the normalized tail moment arm $x^{GN_{ht}}$. According to Quabeck the tail downwind factor $\left(1 - \frac{\partial \epsilon_{ht}}{\partial \alpha_w}\right)$ can be estimated by equation (38) [15]:

$$\frac{\partial \epsilon_{ht}}{\partial \alpha_w} = 1 - \left(\frac{\pi}{4}\right)^2 \cdot \frac{2}{\sqrt{\Lambda_w^2 + 4 + 2}} \quad (38)$$

By entering equations (36) and (37) equation (35) can be solved for tail moment arm $x^{GN_{ht}}$.

$$\frac{x^{GN_{ht}}}{c_{MAC}} = \frac{-c_{L\alpha,w} \cdot \bar{x}^{GN_w} - SM \left(c_{L\alpha,w} + \frac{S_{ht}}{S_{ref}} c_{L\alpha,ht} \right)}{\frac{S_{ht}}{S_{ref}} c_{L\alpha,ht}} \quad (39)$$

Finally, the tail moment arm is iteratively adjusted using PAWAT (see section 2) to counter loss of accuracy by the simplifications in this method as well as downwash on the horizontal tail.

3. Sizing of vertical tail surface area

Quabeck introduces a metric to quantify static lateral stability similar to longitudinal static margin in equation (35) [15].

$$\sigma_s = \frac{c_{N\beta}}{c_{Y\beta}} \quad (40)$$

$c_{N\beta}$ is the yaw moment derivative and $c_{Y\beta}$ the side force derivative due to sideslip. Quabeck suggests values of $-0,8 \leq \sigma_s \leq -0,5$.

Required vertical tail area is then calculated by:

$$\frac{S_{vt}}{S} = \frac{-c \cdot f_F - c \cdot f_w}{c_{L\alpha,vt} \cdot \left(c - \frac{x^{GN_w}}{s}\right)} \quad (41)$$

with fuselage contribution f_F for slender fuselages:

$$f_F = c_{Y\beta,F} \cdot \frac{S_F}{S} \approx 0,08 \cdot \frac{S_F}{S} \quad (42)$$

S_F the wetted surface area of the fuselage, $c_{Y\beta,F}$ its sideforce derivative due to sideslip. Wing contribution f_w can be estimated by:

$$f_w = c_{D,Airfoil} + c_{L\alpha,w} \cdot v^2 \quad (43)$$

$c_{D,Airfoil}$ is the 2D airfoil drag for the design case, $c_{L\alpha,w}$ the wing's lift curve slope and v the dihedral angle. The factor c combines geometrical relations and static lateral stability.

$$c = \frac{x^G}{s} - \frac{x^{N_w}}{s} - \sigma_s \quad (44)$$

VI. Implementation and data modeling

The implementation in MATLAB [16] follows a highly modular approach: It is strictly separated between a data storage model of all parameters describing the aircraft's configuration and calculation methods, which are separated in independent calculation tools. This methodology is not limited to the design process described in section V but can be adapted to any case where different calculation methods are required to work on the same set of parameters describing an aircraft configuration.

The data model of the configuration contains geometry of an arbitrary number of wings and bodies, aerodynamic polars of the entire configuration, propulsion system data for electric or combustion engine systems, data for system components, a set of reference parameters (wing surface, mass, center of gravity) and a container for tool-specific data to store settings and tool-specific, non-general calculation results.

As calculation tool basically every tool can be connected which is integratable in MATLAB. This involves other MATLAB programs, Simulink models as well as standalone tools controlled through command line commands.

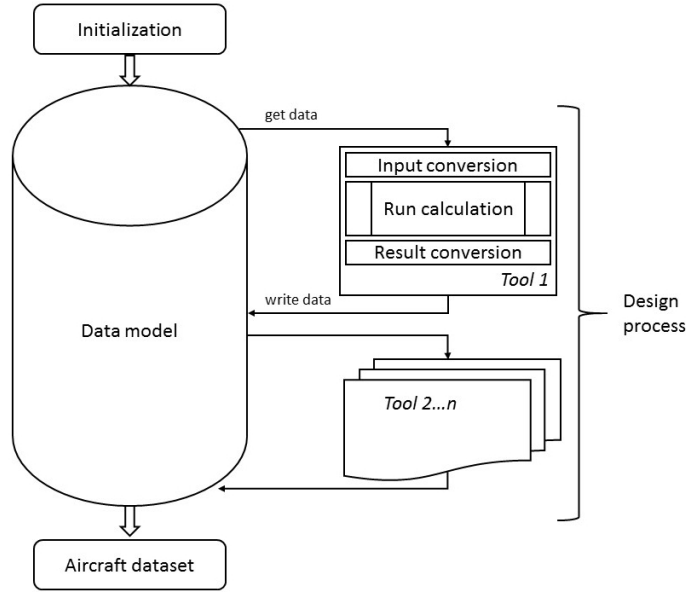


Fig. 10 Implementation of the design process using common dataset

VII. Design study: UAV for mapping

A design study for an unmanned aerial vehicle is carried out. One objective is to determine potential improvement of mission performance by using a stabilized sensor attachment. For the creation of orthomosaic images a certain amount of side overlap between single images of the target area is required. Therefore, side overlap values of 60-65% for unstabilized sensors and only 40% for stabilized gimbals are recommended [3]. By reducing the number of grid lines the flight distance and therefore required propulsive energy and mission time are reduced. On the other hand the aircraft has to be able to carry a heavier sensor payload. This design trade-off is to be addressed by this design study.

A. Description of the design case

Objective of this design case is to create a map of the industrial area of the town Eching and the town of Mintraching, both located north of Munich, Germany. Take-off and landing is performed near the campus of the Technical University of Munich close to Garching.

To achieve acceptable image resolution a Ground Sample Distance ("GSD") value of 5cm is recommended [3].

A Sony alpha 6000 electro-optical camera is to be used as payload, which is very common for this type of task. Table 2 gives an overview of its technical data. Two instances of this sensor were implemented, one with an increased weight of 400grams and slightly increased dimensions to account for stabilization.

Sensor resolution	6000 x 4000	6000 x 4000
Focal length	24mm	24mm
Field of view	53°x 35°	53°x 35°
Dimensions	67 x 45 x 12mm	87 x 55 x 18mm
Weight (camera body & lens)	560g (camera body & lens)	960g (camera body & lens & gimbal)
Gimbal index	1	2

Table 2 Technical data of the camera [17]

As basic configuration a conventional fixed-wing with V-tail was chosen. Wing area S_{ref} , wing aspect ratio Λ , airspeed during the mission $V_{mission}$ and camera index were chosen as design variables for the optimization algorithm.

Objective function is to maximize mission performance index $MPI_{mapping}$ (see equation (45)), and a minimum stall speed of $V_{stall} = 15 \frac{m}{s}$ was chosen to ensure the result shows acceptable handling characteristics.

B. Mission evaluation metric

The mission performance metric used in this case is given by equation (45).

$$MPI_{mapping} = \alpha \cdot \frac{r_{coverage} \cdot f_{deg}}{r_{coverage,ref}} + \beta \frac{E_{Mission,ref}}{E_{Mission}} \quad (45)$$

$r_{coverage}$ is the average value of area coverage rate, $E_{mission}$ the electric energy required to conduct the mission. $r_{coverage}$ is defined as the average value of covered area per time.

$$r_{coverage} = \frac{1}{T} \int_0^T w_{swath} \cdot V dt \quad (46)$$

w_{swath} is the width of the camera footprint on the ground minus required overlap of images.

C. Result

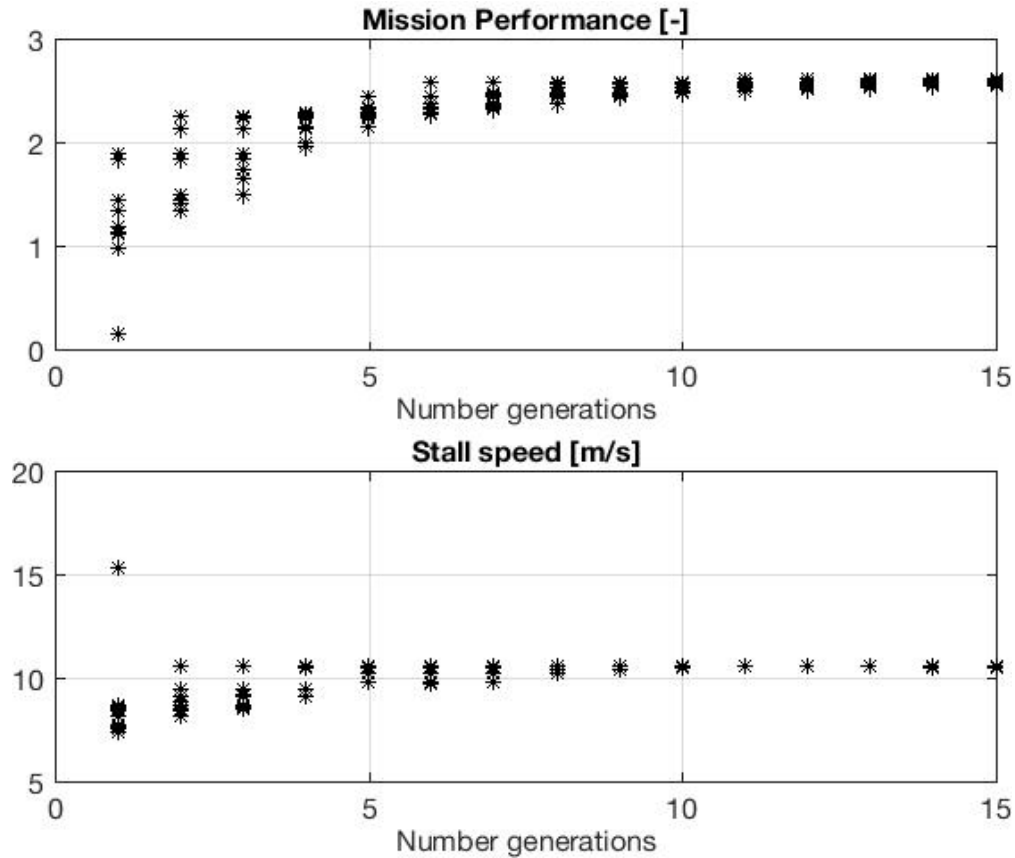


Fig. 11 Progress of mission performance and stall speed during optimization

Figure 11 shows the progress of the objective of maximizing mission performance during the optimization process. Mission performance converges to a value of $MPI_{mapping} = 2.637$, while the restriction of stall speed stays well below the requirement of $V_{stall} \leq 15 \frac{m}{s}$.

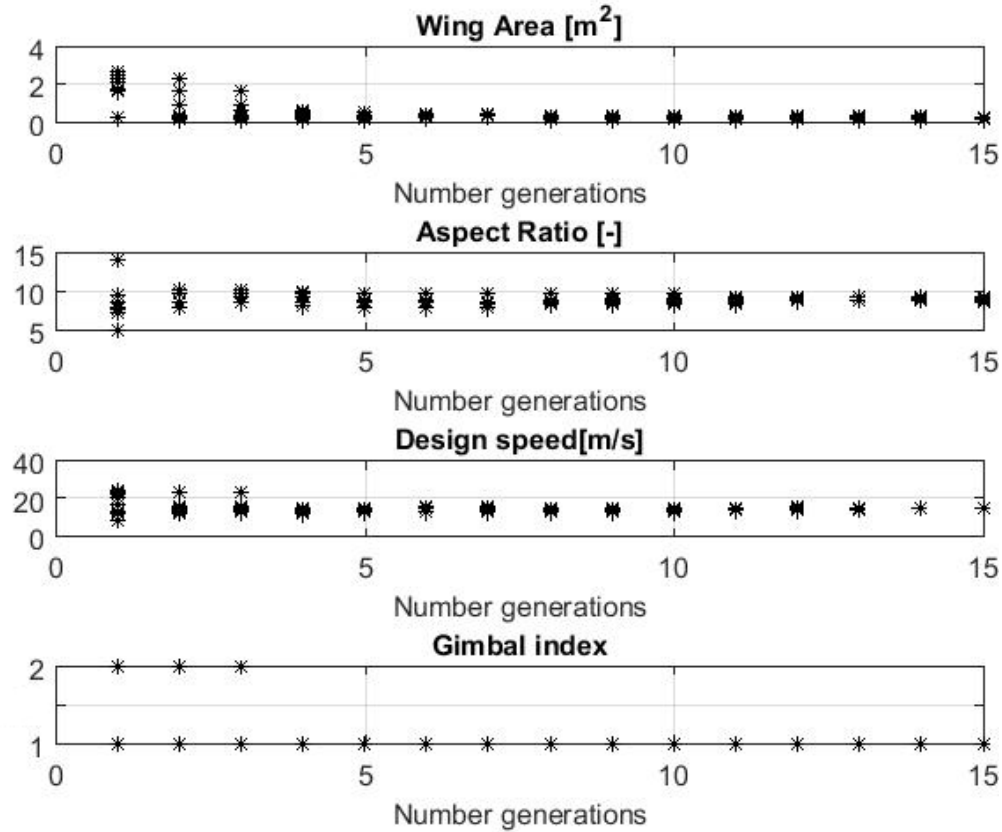


Fig. 12 Progress of design variables during optimization

Figure 12 shows the progress of the design variables during the optimization process. Wing area S_{ref} converges to a value of $S_{ref} = 0.2m^2$, aspect ratio $\Lambda = 8.98$ and design speed $V_{design} = 15.34 \frac{m}{s}$. The last subplot shows the evolution of whether an unstabilized (gimbal index = 1) or a stabilized camera (gimbal index = 2) was chosen by the genetic optimization algorithm. From generation three on only the unstabilized version of the camera was taken into account.

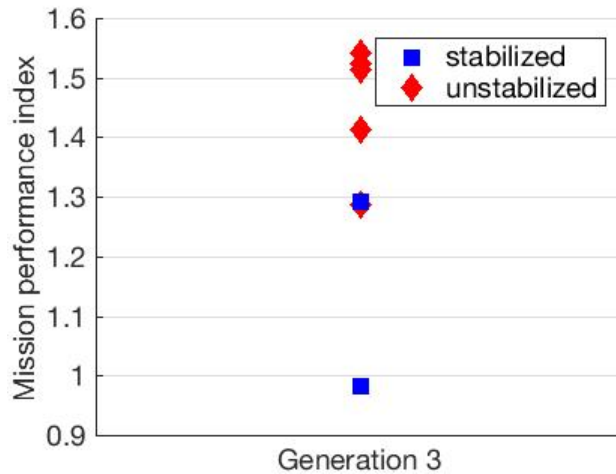


Fig. 13 Comparison of mission performance index for individuals of generation 3

Figure 13 shows the values of $MPI_{mapping}$ for all configurations of generation number three. In this case the difference between the best individuals of either payload type is $1.54 - 1.29 = 0.25$, which is an increase in mission performance of 20%.

Basic parameters of the resulting aircraft configuration are given in table 3, figure 14 shows a sketch of the configuration.

Dimensions	
Wing span	1.5m
Wing area	0.2m ²
Wing aspect ratio	11.33
Weights	
Total weight	1.72kg
Structure weight	0.45kg
Propulsion system weight	0.5kg
Payload weight	0.57kg
Components weight	0.2kg

Table 3 Technical data of the camera [17]

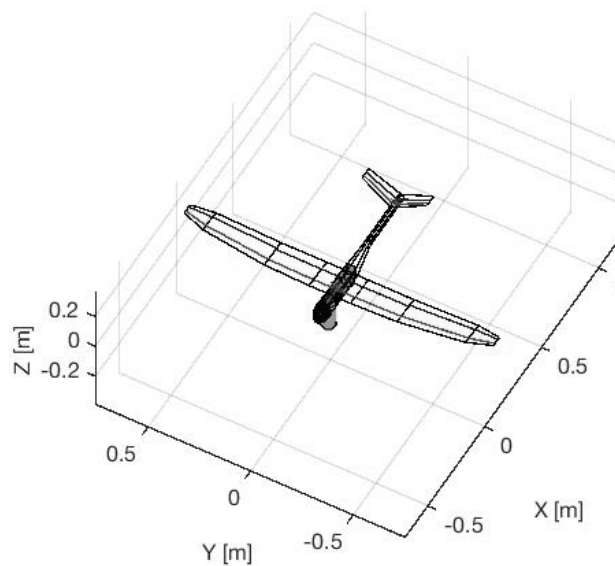


Fig. 14 Sketch of the result configuration

VIII. Conclusion

Since typical missions of Unmanned Aerial Vehicles (UAV) rely on the interaction of the aircraft and sensor payloads, conventional methods for preliminary aircraft design need to be enhanced to take mission specific requirements into account.

To address this necessity a numerical design method for automatically creating consistent designs of UAVs was introduced. To evaluate the mission performance of these designs an approach to create quantifiable mission performance metrics was shown. Based on such a metric, multiple UAV designs can be compared and ranked with respect to their mission effectiveness.

By combining the design process and the mission performance metrics with the genetic optimization algorithm "GAME" an integrated design process was introduced, which automatically generates UAV designs specifically tailored to their

mission requirements .

The method is able to find a well-suited combination of payload sensor and matching aircraft. It enables its user to perform trade-off studies by comparing different payload sensors and aircraft configurations in terms of their mission effectiveness.

References

- [1] Langer, H., "Extended Evolutionary Algorithms for Multiobjective and Discrete Design Optimization of Structures," Dissertation, Technical University of Munich, Munich, Germany, 2005.
- [2] Fokina, E., Feger, J., and Hornung, M., "Application of a visualization environment for the mission performance evaluation of civilian UAS," Aerospace Europe Council of European Aerospace Societies Conference, Bukarest, Romania, 2017.
- [3] sUAS News, "How a Sensor Gimbal Improves Efficiency of an Aerial Mapping Mission," , 2017. URL <https://www.suasnews.com/2017/01/sensor-gimbal-improves-efficiency-aerial-mapping-mission/#print>, online; accessed 17-January-2017.
- [4] Lebourgeois, V., Bégué, A., Labbé, S., Mallavan, B., Prévot, L., and Roux, B., "Can Commercial Digital Cameras Be Used as Multispectral Sensors? A Crop Monitoring Test," *Sensors*, Vol. 8, No. 11, 2008, p. 7300. doi:10.3390/s8117300, URL <http://www.mdpi.com/1424-8220/8/11/7300>.
- [5] Johnson, J., "Analysis of image forming systems," *Proceedings of Image Intensifier Symposium*, Ft. Belvoir, Virginia, 1958, pp. 249–273.
- [6] Gundlach, J., *Designing Unmanned Aircraft Systems: A Comprehensive Approach (Aiaa Education Series)*, AIAA, 2011.
- [7] Schmieder, D., and Weathersby, M., "Detection Performance in Clutter with Variable Resolution," *Aerospace and Electronic Systems, IEEE Transactions on*, Vol. AES-19, No. 4, 1983, pp. 622–630. doi:10.1109/TAES.1983.309351.
- [8] Steiner, H.-J., *Preliminary Design Tool for Propeller-Wing Aerodynamics, Part I: Implementation and Reference Manual*, Bauhaus Luftfahrt e.V., 2010.
- [9] Gur, O., Mason, W., and Schetz, J., "Full Configuration Drag Estimation," San Antonio, Texas, United States of America, 2009.
- [10] Rößler, C., *Conceptual Design of Unmanned Aircraft with Fuel Cell Propulsion System*, Verlag Dr. Hut, 2012.
- [11] Dolch, S., *Rippenflügel aus Faserverbundwerkstoffen*, Verlag für Technik und Handwerk, Baden-Baden, Germany, 1996.
- [12] Schröder, A., "Globale Sichtbarkeitsalgorithmen," Dissertation, Philipps-Universität Marburg, 2003.
- [13] Burns, M., "The STL Format," , 1993. URL http://www.fabbers.com/tech/STL_Format, online; accessed 18-August-2016.
- [14] Feger, J., and Hornung, M., "Numerische Anordnung von Komponenten und Erzeugung von Rumpfgeometrien unbemannter Fluggeräte im Vorentwurfsstadium," *Deutscher Luft- und Raumfahrtkongress*, Munich, Germany, 2017.
- [15] Quabeck, H., *Design, Leistung und Dynamik von Segelflugmodellen*, HQ Modellflugliteratur, Babenhausen, Germany, 1994.
- [16] The Mathworks, I., "MATLAB r2017a," , 2017.
- [17] of America, S. C., "Sony alpha 6000 Full Specifications and Features," , 2017. URL <https://www.sony.com/electronics/interchangeable-lens-cameras/ilce-6000-body-kit/specifications>, online; accessed 05-December-2017.
- [18] Shaw, G. A., and Burke, H. K., *Spectral Imaging for Remote Sensing*, Vol. 14, Lincoln Laboratory Journal, 2003.
- [19] Donohue, J., "Introductory review of target discrimination criteria," Tech. Rep. AD-A254 954, Phillips Laboratory, Air Force Systems Command, Hanscom Air Force Base, Massachusetts, 1991.
- [20] Herbst, S., and Hornung, M., "ADDAM: An Object Oriented Data Model for an Aircraft Design Environment in MATLAB," Tech. Rep. 10.2514/6.2015-3243, AIAA Modeling and Simulation Technologies Conference, testadress, 2015.
- [21] BlackSwift Technologies LLC, "Surveying and Mapping with Fixed Wing UAS, A Fly-Off Between Leading UAS Providers," , 2016. URL <http://www.uasvision.com/2016/11/22/fly-off-compares-aerial-survey-drones/#more-46816>, online; accessed 22-November-2016.

- [22] Nocedal, J., and Wright, S., *Numerical Optimization*, Springer-Verlag New York, 2006.
- [23] Keidel, B., "Auslegung und Simulation von hochfliegenden, dauerhaft stationierbaren Solardrohnen," Dissertation, Technische Universität München, München, 2000.
- [24] Keller, H., Pferschy, U., and Pisinger, D., *Knapsack Problems*, Springer-Verlag, 2004.
- [25] Fahlstrom, P., and Gleason, T., *Introduction to UAV Systems*, UAV Systems Inc., 1998.
- [26] Pamadi, B., *Performance, Stability, Dynamics, and Control of Airplanes*, AIAA education series, American Institute of Aeronautics & Astronautics, 2004.
- [27] Bundesrepublik Deutschland, "SAR-Handbuch Such und Rettungsdienst für Luftfahrzeuge," , 2006. URL <http://www.luftwaffe.de/resource/resource/MzEzNTM4MmUzMzMyMmUzMTM1MzMyZTM2MzIzMzMwMzAzMDMwMzAzMDY3NmY2Njc1NjI2ZTYyNmYyMDIwSAR-Handbuch.pdf>.
- [28] Strohmayr, A., and Schmitt, D., "Scenario based aircraft design evaluation," ICAS 2000 Congress, Harrogate, UK, 2000.
- [29] Mavris, D., and DeLaurentis, D., "An integrated approach to military aircraft selection and concept evaluation," 1st AIAA Aircraft Engineering, Technology, and Operations Congress, Los Angeles, CA, 1995.
- [30] Mavris, D., DeLaurentis, D., Bandte, O., and Hale, M., "A Stochastic Approach to Multi-disciplinary Aircraft Analysis and Design," 36th Aerospace Sciences Meeting and Exhibit, Reno, NV, 1998.